Networking early modern Irish women

近代早期爱尔兰女性的网络化研究

EVAN BOURKE*
Maynooth University
梅努斯大学

ABSTRACT

Over the last decade, network analysis has developed as an approach within digital humanities as a wider array of tools has become available to humanities scholars, and these approaches are now beginning to make an impact on the disciplines of history and English. This article presents an overview of different ways of approaching network analysis. It assesses recent projects to see how they accounted for gender in their datasets and what can be learnt about early modern women from these projects. It then looks at how projects in Ireland are engaging with network analysis, discussing the approaches used by RECIRC and introducing MACMORRIS's analysis of the Dictionary of Irish biography (D.I.B.) and the Bardic Poetry Database (B.P.D.), looking at how the latter is attempting to overcome the unconscious gender bias inherent in the D.I.B.'s selection of early modern lives from the period between 1541 and 1660. Finally, it points to some of the wider issues we as scholars face when engaging with this methodology, such as access to the required training and collaboration, arguing that while these are not unique challenges to the study of gender history in Ireland, they are important debates that can enhance scholarship in the field.

摘要

过去十年间，随着越来越多工具向人文学者开放，网络分析逐渐成为数字人文中的一种重要方法，并开始对历史学和英语文学研究产生影响。本文概述了开展网络分析的不同路径，考察近期项目如何在数据集中处理性别问题，以及这些项目能够让我们了解近代早期女性的哪些方面。随后，文章转向爱尔兰语境，讨论RECIRC所采用的方法，并介绍MACMORRIS对《爱尔兰人物传记辞典》（D.I.B.）和Bardic Poetry Database（B.P.D.）的分析，尤其关注后者如何试图克服D.I.B.在1541至1660年近代早期人物遴选中所内含的无意识性别偏见。最后，文章指出学者采用这种方法时面对的一些更广泛问题，例如获得必要培训与协作支持的机会。虽然这些挑战并非爱尔兰性别史研究所独有，但围绕它们展开的讨论对于推动这一领域的学术研究具有重要意义。

Network analysis — the study of patterns of interconnectedness among a set of things — is a potentially useful method for those studying the lives of early modern women, as it can bring to the fore the agency of lesser-known actors, help raise new questions and open new avenues of research. In Networks: an introduction, Mark E. Newman describes a network as a set of relationships between different objects.1 When visualising a network, the objects are usually referred to as nodes, and the relationships are usually known as edges. For example, the internet could be described as a network, and in this case the nodes would be the device (computer, tablet, mobile phone, for example) and the edges would be the wireless connections between the devices. In historical networks, people tend to be the nodes while their relationships to each other are the edges. These nodes and edges can also convey more than one piece of information, or metadata, in the form of what are known as attributes. In a correspondence network, the nodes contain information on the person, such as their name, date of birth, date of death, sex, religion and role/occupation. The edges contain the letter information, viz. who wrote the letter, to whom was it sent, when the letter was written, where it was sent from and so forth. The attributes in any given network are dictated by the research question, scholarly conventions, the source material, specialist knowledge and quite often the timescale of the project.2 These attributes are important in that they offer different ways to filter and interrogate the network and so they can influence the way the network is conceived and abstracted, its structure, and how this structure is calculated using different network measurements.

网络分析——即研究一组对象之间相互联系的模式——对于研究近代早期女性生活的学者来说，是一种很有潜力的方法，因为它能够把较少受到关注的行动者的能动性凸显出来，帮助提出新的问题，并打开新的研究路径。Mark E. Newman在《网络：导论》（Networks: an introduction）中把网络描述为不同对象之间关系的集合。1 在网络

可视化中，这些对象通常被称为“节点”（nodes），对象之间的关系则被称为“边”（edges）。例如，互联网可以被视为一个网络，其中节点是各种设备（如电脑、平板电脑、手机），边则是设备之间的无线连接。在历史网络中，人通常构成节点，而人与人之间的关系则构成边。节点和边还可以承载不止一项信息，也就是元数据，这些信息以所谓“属性”（attributes）的形式存在。以通信网络为例，节点可以包含人物姓名、出生日期、死亡日期、性别、宗教和身份/职业等信息；边则包含书信本身的信息，例如谁写了信、寄给谁、写于何时、从哪里寄出等等。一个具体网络中应包含哪些属性，取决于研究问题、学术惯例、史料本身、专业知识，而且往往也取决于项目的时间尺度。2 这些属性十分重要，因为它们为筛选和检视网络提供了不同方式，从而能够影响研究者如何理解和抽象化网络、如何构造其结构，以及如何用不同的网络测量方法计算这一结构。

I

一

Scholars such as Newman, Albert-László Barabási, Duncan J. Watts and Linton C. Freeman have analysed a wide range of real-world networks, arguing that they have an underlying order and can be analysed using mathematical tools and models.3 These tools and models have long been used in various disciplines, such as sociology, mathematics and physics, but are beginning to make an impact on the disciplines of history and English.4 While the tools are beginning to be used, this does not mean that the concepts are new to the field. In their recent book, Ruth Ahnert, Sebastian Ahnert, Catherine Nicole Coleman and Scott B. Weingart argue that networks are an abstraction of a concept into a system of nodes and edges, and that these abstract systems are inherently intuitive and are often initially conceived metaphorically — a conceptual framework, they argue, that scholars in the arts and humanities already possess.5 This is a point also picked up by Catherine Medici, who has highlighted that as networks were an integral part of the early modern world, ‘work on political and patronage networks is central to historical, art historical, and literary scholarship’.6 In terms of women’s networks she points to the work of Susan Broomhall and Stephanie Tarbin, Bernard Capp, Amanda Herbert, James Daybell, Julie Crawford, Micheline White and Julie Campbell among others.7

Newman、Albert-L á szl ó Barab á si、Duncan J. Watts和Linton C. Freeman等学者分析了大量现实世界中的网络，并认为这些网络存在某种内在秩序，可以借助数学工具和模型来分析。3
这些工具和模型长期以来已经应用于社会学、数学和物理学等学科，如今也开始影响历史学和英语文学研究。4 不过，工具开始被使用，并不意味着“网络”这一概念对于这些领域来说是新的。Ruth Ahnert、Sebastian Ahnert、Catherine Nicole Coleman和Scott B. Weingart在最近的一本书中指出，网络是把一个概念抽象成由节点和边构成的系统；这种抽象系统本身具有直观性，而且最初往往就是以隐喻方式被构想出来的。他们认为，艺术与人文学者其实早已拥有这种概念框架。5 Catherine Medici也强调，网络本就是近代早期世界的重要组成部分，因此“政治网络和赞助网络的研究，是历史学、艺术史和文学研究的核心内容”。6 在女性网络方面，她列举了Susan Broomhall与Stephanie Tarbin、Bernard Capp、Amanda Herbert、James Daybell、Julie Crawford、Micheline White和Julie Campbell等人的研究。7

In the Irish context, the starting point has to be Margaret MacCurtain and Mary O’Dowd’s edited collection Women in early modern Ireland, which includes important work on women’s political, cultural, religious and kinship networks by scholars such as Bernadette Cunningham, Phil Kilroy, Katharine Simms, Jerrold Casway and Ciarán Brady.8 Consideration of early modern women’s kinship and political networks was also central to the ‘Agenda for women’s history in Ireland’, with MacCurtain, O’Dowd and Luddy highlighting the need for more research on ‘the political patronage of aristocratic women in sixteenth-century Ireland’; this included the networks of the wives of lords deputy and the networks of literate women ‘from wealthy families such as the Butlers of Kilkenny and the Boyles of County Cork’.9 Since its publication there has been an impressive response to this call.10 Both Jerrold Casway and Marie-Louise Coolahan have analysed the political and kinship networks of Rosa O’Dogherty (Róis Ní Dhochartaigh), while Karen Holland has examined the networks of Joan FitzGerald, countess of Desmond.11 The networks of the wives of some lords deputy have been explored by Deirdre Fennell and Catherine Medici, who surveyed the political and kinship networks of the Fitzwilliam women and the Sidney women respectively.12 The networks of the Butlers of Kilkenny and the Boyles of County Cork have received the most sustained analysis since

the publication of the ‘Agenda’. Coolahan, Damien Duffy, Naomi McAreavey and Eleanor O’Keefe have explored the lives of the Butler women; while Coolahan, Ruth Connolly, Michelle DiMeo, Clodagh Tait, Betsy Taylor Fitzsimons, Carol Pal, Ann-Maria Walsh, Ramona Wray and Amelia Zurcher have analysed the lives of the Cork Boyle women, with the most sustained attention being given to the political, religious, cultural and kinship networks of Katherine Jones (née Boyle), Lady Ranelagh and Mary Rich (née Boyle), countess of Warwick.13 The vast majority of these studies have grappled with the concepts of networks in a metaphorical way, employing specialist knowledge and the array of important skills traditionally in a historian’s toolkit — archival research, palaeography, close reading and contextual reading.14 The demands of painstaking, recovery-based research means that the work has largely been restricted to case studies, which by their nature can be relatively limited in size or scope. Now, the development of new digital tools has the potential to widen this scope and network analysis can be incorporated as a complementary element in the work of historians and literary scholars.

在爱尔兰语境中，起点必须是Margaret MacCurtain与Mary O ’ Dowd主编的《近代早期爱尔兰女性》（Women in early modern Ireland）。这部论文集收入了Bernadette Cunningham、Phil Kilroy、Katharine Simms、Jerrold Casway和Ciar á n Brady等学者关于女性政治、文化、宗教和亲属网络的重要研究。8
近代早期女性的亲属网络与政治网络，也是《爱尔兰女性史研究议程》（Agenda for women ’ s history in Ireland）中的核心问题。MacCurtain、O ’ Dowd和Luddy特别指出，需要进一步研究“ 16世纪爱尔兰贵族女性的政治赞助 ”，其中包括总督夫人的网络，以及“ 来自Kilkenny的Butler家族和Cork郡Boyle家族等富裕家庭的识字女性 ”的网络。9 自该文发表以来，这一倡议得到了十分积极的回应。10 Jerrold Casway和Marie-Louise Coolahan都研究过Rosa O ’ Dogherty（R ó is N í Dhochartaigh）的政治与亲属网络，Karen Holland则考察了Desmond伯爵夫人Joan FitzGerald的网络。11 一些总督夫人的网络也得到研究：Deirdre Fennell与Catherine Medici分别考察了Fitzwilliam家族女性和Sidney家族女性的政治与亲属网络。12
自《研究议程》发表以来，Kilkenny的Butler家族与Cork郡Boyle家族受到的持续关注尤其多。Coolahan、Damien Duffy、Naomi McAreavey和Eleanor O ’ Keefe研究了Butler家族女性；Coolahan、Ruth Connolly、Michelle DiMeo、Clodagh Tait、Betsy Taylor Fitzsimons、Carol Pal、Ann-Maria Walsh、Ramona Wray和Amelia Zurcher则研究了Cork的Boyle家族女性，其中受到最多关注的是Katherine Jones（n é e Boyle，Lady Ranelagh）和Mary Rich（n é e Boyle，Warwick伯爵夫人）的政治、宗教、文化与亲属网络。13 这些研究中的绝大多数，主要是在隐喻意义上处理“ 网络 ”概念，并依靠历史学家传统工具箱中的专业知识与技能——档案研究、古文字学、细读和语境化阅读。14 由于这种“ 恢复性研究 ”需要投入大量细致工作，相关研究通常只能集中于个案，其规模和范围也因此相对有限。如今，新型数字工具的发展有潜力扩大这一研究范围，网络分析也可以作为历史学家和文学研究者工作的补充性方法。

II

二

As noted, one of the most beneficial aspects of network visualisation and analysis is the ability to interrogate a network using a variety of network measurements. One such measure is degree centrality, which measures the total number of edges connected to a particular node or, in the case of a social network, the total number of separate people to whom a particular person has connections. People that hold a high degree centrality in networks can be described as hubs, as they have the most connections. Degree centrality is a useful measurement as it can be broken down into ‘in-degree’ and ‘out-degree’. As part of the RECIRC project (a project run by Marie-Louise Coolahan that researched and analysed the ways in which women’s texts were read and transmitted during the early modern period), network visualisation was used to create reception networks that incorporated both of these measures.15 In a reception network, in-degree measures how many different people were writing about a given female author. Out-degree, in contrast, measures how many different female authors a given person was writing about. For example, if you examine RECIRC’s network of receptions that occurs in correspondence, Mary Percy (1570–1642), abbess of the Benedictine monastery in Brussels, is a hub and has both the highest in-degree (i.e. she was being discussed in the letters of various different people) and out-degree (i.e. she was writing about many different women in her letters). This is unsurprising, as in her role as abbess she was involved in a controversy that resulted in her

writing about many of her fellow nuns and they in turn were writing about her.16

如前所述，网络可视化和网络分析最有价值的一点，是能够利用多种网络测量方法来检视一个网络。其中一种是“度中心性”（degree centrality），它衡量与某个特定节点相连的边的总数；在社会网络中，也就是某个人与多少不同的人存在联系。网络中度中心性高的人可以被称为“枢纽”（hubs），因为他们拥有最多连接。度中心性还可以进一步分为“入度”（in-degree）和“出度”（out-degree），因此尤其有用。RECIRC项目由Marie-Louise Coolahan主持，研究并分析近代早期女性文本如何被阅读和传播；该项目利用网络可视化建立“接受网络”，同时纳入了入度和出度两种指标。15 在接受网络中，入度衡量有多少不同的人写到某一位女性作者；出度则相反，衡量某一个人写到了多少位不同的女性作者。例如，在RECIRC基于书信材料建立的接受网络中，布鲁塞尔本笃会修道院院长Mary Percy（1570 – 1642）是一个枢纽节点，她同时拥有最高的入度（即许多不同的人在信中讨论她）和最高的出度（即她在自己的书信中谈到许多不同的女性）。这并不令人意外，因为她作为院长曾卷入一场争议，这使她写到许多同院修女，而这些修女也反过来写到她。16

Another measurement that is important to the examination of networks is eigenvector centrality. A node that ranks highly in this measurement is one that is connected to other nodes that also rank highly. Stephen Borgatti elaborates on this measurement's importance: 'if a node influences just one other node, who subsequently influences many other nodes (who themselves influence still more others)', this measurement will capture the integral role played by the first node in the chain.17 Thus, in this measurement, a node's own importance is dependent on the importance of its neighbouring nodes. This means that the main hubs of a network often rank highly in this measure, but nodes with a relatively low number of connections could still have a high eigenvector score if these connections are to other important nodes. For example, in my study of women's involvement in the intellectual correspondence network known as the Hartlib circle, Katherine Jones, Lady Ranelagh (1615–91), daughter of the first earl of Cork, ranked highly in this measure. This is because she was socially connected to many of the members of this network who were based in England and Ireland.18 This included Samuel Hartlib himself, alongside John Dury, Dorothy Moore, John Beale, William Petty and Benjamin Worsley among others. A final measurement of note is that of 'betweenness' which examines a node's ability to bridge diverse parts of the network. Ruth Ahnert and Sebastian Ahnert note that 'for any two nodes in a network, there is a shortest path between them, and betweenness tells us how many of these shortest paths go through a given node'.19 In their study on Protestant correspondence networks, they found that this measure brought attention to the role played by Anne Smith, wife of the martyr Robert Smith (d. 1555), within the wider correspondence network.20 All of these measures can be invaluable in researching early modern women's lives as, in different ways, they can help reveal the agency of women within larger networks.

研究网络时另一个重要指标是“特征向量中心性”（eigenvector centrality）。一个节点在这一指标中得分较高，意味着它连接到的其他节点本身也具有较高的重要性。Stephen Borgatti这样说明这一指标的重要性：“如果一个节点只影响另一个节点，而后者又影响许多其他节点（这些节点又继续影响更多节点）”，那么这一测量仍然能够捕捉链条中第一个节点所发挥的关键作用。17 因此，在这一指标下，某个节点自身的重要性取决于其相邻节点的重要性。网络中的主要枢纽往往会在特征向量中心性上得分很高，但即便一个节点连接数量不多，只要它连接的是其他重要节点，也可能取得很高的特征向量得分。例如，在我对Hartlib圈这一知识通信网络中女性参与情况的研究里，第一代Cork伯爵之女Katherine Jones、Lady Ranelagh（1615 – 91）在这一指标上排名很高，因为她与许多居住在英格兰和爱尔兰的网络成员都有社会联系。18 这些人包括Samuel Hartlib本人，以及John Dury、Dorothy Moore、John Beale、William Petty和Benjamin Worsley等。另一个值得注意的指标是“中介中心性”（betweenness），它考察一个节点连接网络不同部分的能力。Ruth Ahnert和Sebastian Ahnert指出，“网络中任意两个节点之间都存在一条最短路径，而中介中心性告诉我们，有多少这样的最短路径会经过某个给定节点”。19
他们在研究新教徒通信网络时发现，这一指标把人们的注意力引向了殉道者Robert Smith（卒于1555年）的妻子Anne Smith在更大通信网络中的作用。20 对于研究近代早期女性生活来说，这些指标都很有价值，因为它们可以从不同角度揭示女性在更大网络中的能动性。

While network visualisation is a useful tool for the analysis of various forms of historical documents, it is especially useful in the analysis of correspondence. Gary Schneider describes early modern

letters as ‘sociotexts’: they demonstrate ‘material evidence of social connectedness’. Thus, social network visualisation is an effective way of highlighting and bringing attention to the social ties that letters ‘initiated, negotiated and consolidated’.21 Two scholars to pioneer this approach with regard to correspondence are Ahnert and Ahnert.22 In their first study, they used quantitative network analysis to visualise and analyse the Protestant letter network memorialised in John Foxe’s Acts and monuments, or the Book of Martyrs as it is more popularly known. In order to do this, they analysed 289 letters ‘that were written either by or to Protestants residing in England during Mary’s reign’, finding that female sustainers, such as Joyce Hales and Margery Cooke, played a more central role than had previously been suspected.23 In their second project, Tudor Networks of Power (TNOP), Ahnert and Ahnert analysed 132,747 letters in the state papers, from the accession of Henry VIII in 1509 until the death of Elizabeth I in 1603, using metadata granted to them by State Papers Online (SPO). They note that given the political nature of the archive, women only make up 4 per cent of their corpus. However, they stress that, as their study examines 20,656 correspondents, this four percent actually accounts for around 826 distinct women, highlighting the importance of this source for the recovery of women’s lives and political activity.24

虽然网络可视化可以用于分析多种历史文献，但它尤其适合书信材料。Gary Schneider把近代早期书信称为“社会文本”（sociotexts），认为它们体现了“社会连接的物质证据”。因此，社会网络可视化能够有效突出书信所“开启、协商并巩固”的社会联系。21 Ahnert和Ahnert是较早把这种方法应用到书信研究中的学者。22 在第一项研究中，他们使用定量网络分析，对John Foxe《Acts and Monuments》（更广为人知的名称是《殉道者之书》）中保存的新教徒书信网络进行可视化和分析。他们分析了289封“由Mary统治时期居住在英格兰的新教徒写出或写给他们的书信”，发现Joyce Hales、Margery Cooke等女性支持者在网络中的中心作用，比此前人们想象的更大。23 在第二个项目“Tudor Networks of Power（TNOP）”中，Ahnert和Ahnert使用State Papers Online（SPO）提供的元数据，分析了从1509年Henry VIII即位到1603年Elizabeth I去世之间国家档案中的132,747封书信。他们指出，由于这一档案具有强烈的政治性质，女性只占语料的4%。但他们也强调，整个研究共涉及20,656名通信者，因此这4%实际代表约826名不同女性，这说明该史料对于恢复女性生活及其政治活动具有重要意义。24

Ahnert and Ahnert’s study provides a foundation for further research, allowing scholars to take a perspectivist approach (a network with a particular person placed at the centre) facilitating an examination of the networks of these 826 women. From an Irish perspective, this includes the political networks of Joan FitzGerald, countess of Desmond (d. c.1595); Elinor FitzGerald, countess of Desmond (c.1545–c.1638); Mabel FitzGerald, countess of Kildare (c.1536–1610); and Anne Fitzwilliam (d. 1602).25 There are also several other studies built around correspondence networks that focus on gender as a significant lens of analysis. This includes work by Catherine Medici, who is working on the Sidney family letters in order to analyse the central role of women in the family’s political and literary networks; Bronagh Ann McShane, who examined letters written by and about English Benedictine nuns living in Brussels, which shows how the enclosure of the nuns was compromised by lay relatives both in and beyond Spanish Flanders; and my own work which analyses Katherine Jones, Lady Ranelagh and Dorothy Moore’s position in the Hartlib circle.26

Ahnert与Ahnert的研究为进一步工作提供了基础，使学者可以采用一种“视角化”方法——把某个特定人物放在网络中心——来考察这826名女性各自的网络。从爱尔兰角度看，其中包括Desmond伯爵夫人Joan FitzGerald（约卒于1595年）、Desmond伯爵夫人Elinor FitzGerald（约1545 – 约1638）、Kildare伯爵夫人Mabel FitzGerald（约1536 – 1610）以及Anne Fitzwilliam（卒于1602年）的政治网络。25 还有一些以通信网络为基础的研究，把性别作为重要分析视角。例如，Catherine Medici正在研究Sidney家族书信，以分析女性在该家族政治与文学网络中的核心作用；Bronagh Ann McShane考察了居住在布鲁塞尔的英格兰本笃会修女所写以及关于她们的书信，指出修女的封闭生活如何被西属佛兰德内外的世俗亲属所打破；而我自己的研究，则考察了Katherine Jones、Lady Ranelagh与Dorothy Moore在Hartlib圈中的位置。26

In general, there are three ways of assembling a dataset, as exemplified by a number of existing projects. First, there are hand-curated datasets that draw on digitised corpuses that have incorporated full text transcriptions, such as Medici’s study of the Sidney network, which is based on a digitized version of the Sidney family letters edited by Arthur Collins in 1746, and my study of the

Hartlib circle, which used transcriptions available on Hartlib Papers Online.27 Secondly, there are the datasets that have extracted and created networkable metadata from meta-archives (digitised calendars and registers); for example, Ahnert and Ahnert's Tudor Networks of Power, which used the metadata generated by the digitisation of the calendars of the state papers; and Błoch et al.'s study of official correspondence in the early modern Portuguese empire, which used natural language processing techniques on a digitised register of correspondence created by the Historical Overseas Archive of Lisbon.28 Finally, there is manual transcription and data entry from both print and manuscript sources, as exemplified by Ahnert and Ahnert's study of the correspondence in Acts and monuments and McShane's study of the reception networks of early modern nun's letters, which drew on manual transcriptions and metadata collected and curated for the RECIRC project by both Emilie K. M. Murphy and McShane.29 This highlights two issues that I will return to later — access to digitised material and the labour involved in building these datasets. Other forms of life writing also offer opportunities for this kind of analysis. Medici, for example, notes that Melissa Schultheis is working on building networks from recipe books, specifically looking at Lady Ann Fanshawe's book of receipts in order to extract human and ingredient networks, to explore 'how network analysis helps understand the possibility of a royalist palette'.30

总体而言，现有项目大致展示了三种构建数据集的方式。第一种是人工策划的数据集，建立在已经数字化并包含全文转录的语料库之上。例如，Medici对Sidney网络的研究使用了Arthur Collins于1746年编辑的Sidney家族书信数字版，而我对Hartlib圈的研究则使用Hartlib Papers Online提供的转录文本。27 第二种，是从“元档案”（即数字化的目录、日历和登记册）中抽取并建立可用于网络分析的元数据。例如，Ahnert与Ahnert的Tudor Networks of Power使用了国家档案日历数字化后生成的元数据；Błoch等人研究近代早期葡萄牙帝国的官方通信，则对里斯本Historical Overseas Archive制作的数字化通信登记册使用了自然语言处理技术。28 第三种，是从印刷材料和手稿中进行人工转录和数据录入。例如，Ahnert与Ahnert对《Acts and Monuments》中书信的研究，以及McShane对近代早期修女书信接受网络的研究，均依赖人工转录，以及Emilie K. M. Murphy和McShane为RECIRC项目收集和整理的元数据。29 这也凸显了我稍后还会回到的两个问题：数字化材料是否可获得，以及建立这些数据集需要投入多少劳动。其他形式的自我书写同样可以使用这种分析。Medici指出，Melissa Schultheis正在从食谱/配方簿中建立网络，尤其研究Lady Ann Fanshawe的配方簿，从中抽取人物网络和食材网络，以探讨“网络分析如何帮助我们理解一种保王党口味成为可能的条件”。30

Less work has been done on other genres of life writing, and developing network-analysis methodologies for diaries and autobiographies will widen the questions literary scholars and historians can ask. Diaries would be particularly amenable to this method, in that each entry could be treated as a separate element, allowing for the exploration of kinship and familial relationships and how these change over the diarist's writing life. Once a methodology has been developed for working on diaries and autobiographies, we can start to ask wider questions: Are specific ties (links to family, servants, broader kinship networks) emphasised or elided depending on genre? Does the structure of networks produced by a certain genre of life writing change over time? Scholars of early modern Irish women are well placed to engage in the development of these methodologies and examine what they can tell us about early modern women's lives, especially considering the plethora of life writing linked to the Cork Boyle family. For example, the diary of Mary Rich, which has almost daily entries for a ten-year period of her life (1666–76), is one potential avenue, while an examination of the role of women in the diary of the second earl of Cork is another possibility.

对其他类型自我书写材料的网络研究相对较少。如果能够为日记和自传发展出网络分析方法，文学研究者和历史学家能够提出的问题将会明显扩大。日记尤其适合这种方法，因为每一篇日记条目都可以被视为一个独立元素，由此可以考察亲属关系和家庭关系，以及这些关系如何随着作者持续写作而发生变化。一旦日记和自传的分析方法得到发展，我们就可以提出更广泛的问题：不同文类是否会强调或省略某些特定联系，例如与家庭成员、仆人或更广泛亲属网络的联系？某一种自我书写文类所生成的网络结构，会不会随着时间变化？研究近代早期爱尔兰女性的学者非常适合参与这些方法的开发，并借此重新理解女性生活，尤其是考虑到Cork的Boyle家族留下了大量自我书写材料。例如，Mary Rich的日记在1666至1676年的十年间几乎每日都有记录，是一个很有潜力的研究方向；第二代Cork伯爵日记中女性所扮演的角色，也可以成为另一条路径。

Another useful approach to network analysis/visualisation that has gained traction is the abstraction of networks from biographical sources. One such project is BiographySampo, which drew on 13,100 biographies in the collections of the Biographical Centre of the Finnish Literature Society to create what are known as reference networks. In their case, they built networks from the embedded 'HTML links and mentions of people in the biographies to create a reference network which is analogous to citation networks'. In these networks, 'the nodes are people, and when a person A is mentioned in the biography of B, a directed edge is added from B to A'.31 Once this process was complete, the transformed network data could be used in applications and queried, allowing perspective networks for an individual or a group to be generated automatically on their website. A betterknown project, and more pertinent to early modern studies, is the Six Degrees of Francis Bacon (SDFB) project, which has abstracted a network of early modern England using the Oxford Dictionary of National Biography (O.D.N.B.) as its dataset. This project takes a different approach to BiographySampo: instead of creating reference networks, it uses a statistical method for abstracting the network. First, SDFB extracted names from the biographies using Name Entity Recognition (NER) software (excluding names that occur in fewer than five documents). They then designed and implemented a statistical inference algorithm that determined the likelihood of one person knowing another person based on the information in the O.D.N.B., assigning a confidence score for the connection, before using various validation tests on the network.32

网络分析/可视化中另一个日益受到重视的方法，是从传记资料中抽象出网络。BiographySampo项目就是一个例子。它利用芬兰文学学会传记中心收藏的13,100篇传记建立所谓“引用/指涉网络”（reference networks）。项目通过传记中嵌入的“HTML链接和对人物的提及，建立一种类似引用网络的指涉网络”。在这种网络中，“节点是人物；当人物A在人物B的传记中被提及时，就从B到A添加一条有向边”。31 完成这一转换之后，网络数据便可以在应用中使用和查询，网站还能自动为某个个人或群体生成“视角网络”。另一个更广为人知、也更与近代早期研究相关的项目，是Six Degrees of Francis Bacon（SDFB）。该项目以《Oxford Dictionary of National Biography》（O.D.N.B.）为数据集，抽象出一个近代早期英格兰的社会网络。它与BiographySampo不同，并不是建立指涉网络，而是使用统计方法抽象网络。首先，SDFB用命名实体识别（NER）软件从传记中提取人名，并排除在少于五篇文献中出现的名字；随后设计并实施统计推断算法，根据O.D.N.B.中的信息计算两个人彼此认识的可能性，为这种联系赋予置信度分数，最后再用多种验证测试检查网络。32

The overall aim of the project was to 'represent and reinforce the full diversity of early modern social ties', but they note that historical scarcity and their own editorial decisions prevent this. One such shortcoming is gender, but the SDFB team have noted and addressed this issue in several ways. A figure is present in these networks only if they are already privileged by the wider historiography; absence, therefore, does not indicate non-existence. As networks are abstractions of the data contained in the dataset, someone absent from the dataset cannot be in the network. For SDFB, this led, as they acknowledge, to algorithmic bias against women for two key reasons. The first of these was gender bias within the O.D.N.B. itself where, by the time SDFB was finishing up, only 534 of 9,929 (5.4 per cent) of early modern biographies are of women's lives. The second was where women's names changed with marriage, causing some women to not reach the project's threshold for inclusion.33 Despite these issues, their initial method resulted in 6.6 per cent of the dataset being women, and these women had a median degree (number of people they were connected to) of 20.3 people, while the men in the dataset had a median degree of 25.7. They then followed this up with 'addathon' to manually include women, which drew participants from all over the world to add early modern English women and their connections to the dataset.34 While SDFB's final dataset is not close in terms of gender parity, it is still an important project. Significantly, the team's openness to the biases of data and the impact of absent or missing data allows us to think critically about gender in the sources that they used. Since the close of the project, the O.D.N.B. has commissioned several series of early modern women's lives, adding the biographies of women such as Elizabeth Dowdall (fl. 1640–42), and Frances Devereux (c.1568–1632) to the dataset, thus attempting to address the biases in their corpus.35

这个项目的总体目标，是“呈现并强化近代早期社会联系的全部多样性”，但项目团队也指出，史料的缺失以及他们自己的编辑决定，使这一目标不可能完全实现。性别就是其中一个明显缺陷，不过SDFB团队已经从多个方面注意并处理了这个问题。一个人物只有在更广泛的历史书写中已经获得足够重视，才会出现在这些网络里；因此

，“ 没有出现 ”并不等于“ 不存在 ”。由于网络只是数据集中材料的抽象，如果某个人根本不在数据集中，他就不可能进入网络。SDFB承认，这造成了两方面对女性不利的算法偏差。第一，O.D.N.B.本身就存在性别偏差：到SDFB项目即将结束时，9,929篇近代早期人物传记中只有534篇（5.4%）写的是女性。第二，女性婚后姓名的变化会导致同一女性的记录分散，从而使一些女性达不到项目设定的纳入阈值。33 尽管如此，项目最初的方法仍使女性占数据集的6.6%；女性节点的度数中位数，也就是她们平均连接到多少人，为20.3，而男性节点的中位数为25.7。随后，团队又举办“ addathon ”活动，以人工方式补充女性数据，吸引世界各地参与者把近代早期英格兰女性及其联系加入数据集。34 SDFB最终的数据集距离性别平衡仍很远，但依然是一个重要项目。更重要的是，团队公开讨论数据偏差，以及缺失数据可能造成的影响，使我们能够批判性思考他们所使用史料中的性别问题。项目结束后，O.D.N.B.又委托撰写了多批近代早期女性传记，把Elizabeth Dowdall（活跃于1640 – 42）和Frances Devereux（约1568 – 1632）等女性加入数据集，试图修正原有语料中的偏差。35

III

三

While many of the projects mentioned touch on or include early modern Irish women, there are two key digital humanities projects, both based in Ireland, that are interested in the digital recovery of early modern Irish women’s lives through network analysis and networked approaches.36 The first is MACMORRIS (Mapping Actors and Communities: Modelling Research in Renaissance Ireland in the Sixteenth and Seventeenth Centuries), on which I am currently employed as a project manager; the second is RECIRC (The Reception and Circulation of Early Modern Women’s Writing, 1550–1700), on which I previously worked as a Ph.D. and postdoctoral researcher. Led by Patricia Palmer, MACMORRIS is a four-year Irish Research Council-funded project that aims to map the full range and richness of cultural activity, across languages and ethnic groups, in Ireland, from 1541 to 1660. It seeks to present a revised and expanded picture of early modern Ireland, one that goes against the predominantly Anglocentric perspective of literary-historical scholarship by developing an open-access resource that ‘maps significant cultural actors (of whatever ethnicity) writing in, or engaging with, Gaelic, English, Latin, Scots, Spanish, Italian, Portuguese, and Dutch, in late 16th and early 17th century Ireland’.37 Its focus is on ‘cultural actors’, which are defined as both those who produce culture (primary cultural actors), and those who shape the contexts within which cultural practices operate and change over time (secondary cultural actors). The project is developing two interactive visualisations to determine how far the recovery of such cultural actors — and cultural acts — can capture the layered complexity of cultural conflict and change over time: 1) a network graph of cultural actors and their connections; and 2) a deep map of early modern Munster. In terms of the network graph, MACMORRIS draws on the metadata of two pre-established projects: the Dictionary of Irish Biography (D.I.B.) and the Bardic Poetry Database (BPD).38 The project aims to use this metadata to extract the social connections from the D.I.B., alongside the patronage connections evident in bardic praise poetry contained in the BPD and combine them on the same interoperable network graph. This will allow us to explore how the Gaelic-speaking world of the bardic poets was intertwined with the Anglocentric world of Elizabethan conquest, despite the purposeful absence of the Irish language in the Anglocentric sources and its marginalisation within the predominantly Anglocentric perspective of literary-historical scholarship.

前面提到的许多项目都涉及或包含近代早期爱尔兰女性，但真正以网络分析和网络化方法进行女性数字“ 恢复 ”的关键数字人文项目有两个，而且都位于爱尔兰。36 第一个是MACMORRIS（Mapping Actors and Communities: Modelling Research in Renaissance Ireland in the Sixteenth and Seventeenth Centuries），我目前在其中担任项目经理；第二个是RECIRC（The Reception and Circulation of Early Modern Women ’ s Writing, 1550 – 1700），我此前曾在其中担任博士研究员和博士后研究员。MACMORRIS由Patricia Palmer主持，是一个由Irish Research Council资助、为期四年的项目，目标是绘制1541至1660年爱尔兰跨语言、跨族群文化活动的全部范围与丰富性。它试图修正并扩展我们对近代早期爱尔兰的认识，反驳文学史研究中以英格兰为中心的主导视角，并建设一个开放获取资源，用来“ 绘制16世纪晚期至17世纪早期爱尔兰中，以Gaelic、英语、拉丁语、Scots、西班牙语、意大利语、葡萄牙语和荷兰语写作或与这些语言发生互动的重要文化行动者（不论其族群）”。37 项目的核心概念是“ 文化行动者 ”，既包括直接生产文化的人（一级文化行动者），也包括塑造文化实践运行环境并影响其随时间变化的人（二级文化行动者）。项目正在开发两种互动可视化，以考察对这些文

化行动者——以及文化行为——的恢复，在多大程度上能够呈现文化冲突与变化的层层复杂性：其一是文化行动者及其联系的网络图，其二是近代早期Munster的“深度地图”。在网络图方面，MACMORRIS使用两个既有项目的元数据：《Dictionary of Irish Biography》（D.I.B.）和Bardic Poetry Database（BPD）。38 项目希望利用这些元数据，从D.I.B.中抽取社会联系，同时提取BPD中吟游诗歌赞诗所显示的赞助关系，再把二者放到同一张可互操作的网络图上。这样便可以考察：尽管以英格兰为中心的史料有意排除爱尔兰语，而文学史研究也长期以英格兰中心视角边缘化爱尔兰语世界，吟游诗人的Gaelic语世界究竟如何与伊丽莎白时期征服的英格兰中心世界交织在一起。

The overall aim of MACMORRIS's analysis of the D.I.B. is very similar to SDFB, in its use of the O.D.N.B., but MACMORRIS involves a much smaller dataset. In terms of size, for the period 1540–1660 there are 1,068 biographies in the D.I.B., compared to around 10,000 in the O.D.N.B., and in terms of politics, the 1,068 biographies have been chosen for their importance to Irish rather than British history (as defined by the editors of both projects). Thus, rather than employ the algorithmic approach of SDFB, MACMORRIS has followed a similar method to that of BiographySampo. First, the team extracted a reference network, connecting person B to person A when person A was mentioned in the biography of B; this is a semi-automated process, which involves the use of Named Entity Recognition (NER). We then read through the biographies to manually correct errors and compile biographical metadata on the actors extracted from entries in which they are mentioned. We subsequently extracted a co-citation network in that we connected B to A when person A and person B were both mentioned in the same biography. We are in the process of verifying these potential connections, as while two people mentioned together in a number of different biographies is highly suggestive of a connection between them, it is possible that the co-citation might indicate nothing more than mutual acquaintances.39 In terms of gender, this approach will allow a focus on political, cultural and kinship networks and highlight women who are mentioned in the biographies of men but who themselves often remain on the periphery of scholarship. MACMORRIS faces many of the same challenges as SDFB, in terms of the biases inherent in our source material and the effect this has on the abstracted network. Much like their network, absence in the MACMORRIS dataset does not equal non-existence. Like the O.D.N.B., the D.I.B. has a gender bias in that of the 1,068 early modern lives, only 38 (3.5 per cent) are of women. This represents an even greater bias against women than that found in the O.D.N.B. Our smaller dataset and method of manually reading the biographies to verify the connection extracted by NER meant that we were able to disambiguate and deduplicate women mentioned in biographical entries and did not have to employ a threshold for inclusion. As a result, we were able to attempt to address the gender imbalance; by incorporating women mentioned in the biographies, we have been able to ensure that, at the time of writing, 496 of 3,173 (15.5 per cent) people in the dataset are women and the open-access version of the dataset will include a perspective and interactive network of every early modern Irish woman mentioned in this article to date. Much like the recent efforts by the O.D.N.B. to address biases in their corpus, the difference between the number of women mentioned in D.I.B. biographies and those that have their own entries highlights the need for the D.I.B. to follow suit and expand its corpus of early modern women's lives, especially as an open-access resource that has the potential to reach a diverse audience.

MACMORRIS对D.I.B.的分析目标，与SDFB使用O.D.N.B.时非常相似，但MACMORRIS的数据集小得多。从规模看，1540至1660年间D.I.B.包含1,068篇传记，而O.D.N.B.约有10,000篇；从政治取向看，这1,068篇传记的入选标准，是人物对爱尔兰史而不是英国史的重要性——至少两套项目的编辑者是这样界定的。因此，MACMORRIS没有采用SDFB的算法路径，而采取了更接近BiographySampo的方法。首先，团队抽取指涉网络：如果人物A在人物B的传记中被提到，就把B连接到A。这一过程是半自动的，需要使用命名实体识别（NER）。随后，我们通读传记，人工纠正错误，并为从传记条目中抽取出的行动者补充传记元数据。之后，我们又建立“共被引网络”：当人物A和人物B同时出现在同一篇传记中时，就连接B和A。我们目前仍在核实这些潜在联系，因为两个名字在多篇不同传记中反复同时出现，虽然强烈暗示二者可能存在关系，但这种共现也可能仅仅意味着他们拥有共同熟人。39 从性别角度看，这种方法能够把注意力放到政治、文化和亲属网络上，并突出那些经常出现在男性传记中、但自身长期处于学术研究边缘的女性。MACMORRIS在史料偏差及其如何影响抽象网络方面，面对许多与SDFB相同的问题。与SDFB一样，人物缺席于MACMORRIS数据集，并不意味着其不存在。D.I.B.也像O.D.N.B.一样存在性别偏差：1,068位近代早期人物中，只有38位（3.5%）是女性，这一比例甚至比O.D.N.B.对女性的偏差更严重。由于我们的数据集更小，而且会人工阅读传记来核实NER抽取出的联系，因此能够消除传记中女性名字的歧义和重复，也无需设置一个纳入阈值。这样，我们就可以尝试修正性别失衡：把传记中提及的女性也纳入后，在本文写作时，

数据集3,173人中已有496人（15.5%）为女性；开放获取版本还将为本文迄今提到的每一位近代早期爱尔兰女性提供一个以她为中心的互动网络。就像O.D.N.B.近来努力修正自身语料偏差一样，D.I.B.传记中“被提到的女性”数量与“拥有独立词条的女性”数量之间的差距，也说明D.I.B.需要采取类似做法，扩充近代早期女性人物语料，尤其因为它是一项有潜力触达多元读者的开放获取资源。

The second dataset that MACMORRIS has worked with is the Bardic Poetry Database. Bernadette Cunningham has argued that, despite women's apparent exclusion from the bardic schools, we need to attend to the role of 'women as patrons, women as authors and women as subjects of literature'.40 In our analysis of the BPD, we were aware of the different roles women could play in the production of culture and we wished to highlight the explicit role of women in bardic poetry. This section of the project is led by Deirdre Nic Chárthaigh, who, when manually extracting and cleaning the dataset, marked up the identity of the poet, the identity of the patron, the gender of the poet and patron, and recorded instances where a wife/mother of a patron was praised in dedicated quatrains of a poem.41 Overall, we extracted 720 people from the BPD, of which 105 (14.5 per cent) were women — 71 of whom were written about in praise poems dedicated to their husbands or sons. We then extracted patronage networks and reference networks in order to explore the different roles played by women. This has resulted in us extending the understanding of women's role as patrons: we now know that, in a network where both poet and patron of a particular poem can be identified, 32 of the 269 patrons (12 per cent) are women, including Martha Stafford (d. 1678), Nuala Ní Domhnaill (c.1575–1630), and Onóra inghean Uí Bhriain (c.1569), and that of these 32 women, ten of them acted as a patron for two different poets. While the vast majority of the connections extracted from the D.I.B. and BPD are male to male (which is to be expected, considering the sources we were analysing), MACMORRIS shows the potential of network analysis to recover elements of the lives of women in early modern Ireland when gender as a category for data collection and analysis is given due consideration, even if the wider aims of the analysis are not exclusively related to gender.

MACMORRIS处理的第二个数据集是Bardic Poetry Database。Bernadette Cunningham指出，尽管女性表面上被排除在吟游诗人学校之外，我们仍必须注意“女性作为赞助人、女性作为作者以及女性作为文学对象”的角色。40 在分析BPD时，我们意识到女性可以在文化生产中扮演多种角色，因此希望突出她们在吟游诗歌中的明确作用。这一部分由Deirdre Nic Ch á rthaigh负责。她在人工抽取和清洗数据时，标记诗人身份、赞助人身份、诗人和赞助人的性别，并记录某位赞助人的妻子或母亲是否在一首诗的专门四行诗节中受到赞美。41 我们最终从BPD中提取了720人，其中105人（14.5%）为女性；这105人中有71人是在献给她们丈夫或儿子的颂诗中被写到的。随后，我们又抽取赞助网络和指涉网络，以考察女性所扮演的不同角色。这扩展了我们对女性作为赞助人的认识：在能够同时识别一首诗的诗人和赞助人的网络中，269位赞助人里有32位（12%）是女性，其中包括Martha Stafford（卒于1678年）、Nuala N í Domhnaill（约1575 – 1630）和On ó ra inghean U í Bhriain（约1569）；而这32名女性中，有10人曾同时担任两位不同诗人的赞助人。虽然从D.I.B.和BPD抽取的大多数联系都是男性与男性之间的联系——考虑到我们所分析的史料，这并不意外——但MACMORRIS显示，只要在数据收集和分析阶段认真把性别作为一个类别，即使整个研究并非专门研究性别，网络分析也有潜力恢复近代早期爱尔兰女性生活中的重要片段。

In the case of the second Irish-based project, the RECIRC project, gender is at the heart of its approach to networks. RECIRC (2014–20) was a European Research Council-funded project, run by Marie-Louise Coolahan, which researched and analysed the ways in which women's texts were read and transmitted during the early modern period. RECIRC produced a large-scale, quantitative analysis of the reception and circulation of women's writing from 1550 to 1700, in order to enable analysis of how texts, ideas and reputations gained traction in the early modern period. In its initial phase, the project designed a database for its researchers to store their data in a single online location, using a centralised taxonomy that enabled them to compare findings, even if they were drawn from different archives around the world.42 The final phase of the project designed a new user interface to share these findings with a wider public and to promote the study of both women's writing and its reception history. Coolahan notes that: the open-access version enables users to search almost 5,000 instances of reception evidence, filtered by categories that include date, place, and source, as well as reception type, circulation type, and three categories of person: the female author, the receiver, and the owner/compiler/scribe of the volume in which the instance of reception

occurs.43

第二个以爱尔兰为基地的项目RECIRC，则把性别直接置于网络方法的核心。RECIRC（2014 – 20）由European Research Council资助，由Marie-Louise Coolahan主持，研究并分析近代早期女性文本如何被阅读和传播。项目对1550至1700年女性写作的接受与流通进行了大规模定量分析，以便进一步研究文本、思想和声誉在近代早期如何获得影响力。项目初期设计了一套数据库，让研究者可以把数据集中存储在同一个在线位置，并采用统一分类体系，从而使来自世界各地不同档案馆的发现也能够相互比较。42 项目最后阶段又设计了新的用户界面，把这些成果分享给更广泛的公众，并推动女性写作及其接受史研究。Coolahan指出，开放获取版本允许用户检索近5,000条“接受证据”，并可按日期、地点和来源等类别进行筛选，还可以按接受类型、流通类型以及三类人物进行筛选：女性作者、接受者，以及出现该接受实例的卷册之所有者/编纂者/抄写者。43

The aim of this platform was to empower users to generate their own results and visualisations, provoking this by displaying the results in visual formats, including network graphs. This enabled users interested in early modern Ireland to search and visualise the reception networks of women such as Katherine Philips (1632–64), Anne Southwell (1574?–1636) and Dorothy Moore (1613–64). For example, in the case of Katherine Philips, it was possible to explore visually who was compiling her work, the types of sources they compiled her work in and who was writing about her. Like most humanities data, the networks generated by RECIRC are complex and are all either bimodal or multimodal. This means that they are made up of two or more node types. In these networks, the nodes (or actors) can be one of four types: female author; receiver; owner/compiler/scribe; and reception source type. If the network represented has two of these four types (for example, female author and receiver), this means that the network is bimodal; while if the network has three or more of these types, then it is a multimodal network. Scott Weingart has shown that while humanities data is often rich with various node types (such as people, places and things): ‘the more complex and multimodal your dataset, the less you can reasonably do with it’.44 He argues that in a bimodal network, ‘degree centrality is nothing more than a count of affiliations’, and that ‘the simplest definition of clustering coefficient doesn’t work on bimodal networks’. For users with only a beginner-level understanding of network analysis, these issues are then compounded by the limitations of open-source software, such as the software platform Gephi’s inability to differentiate between unimodal (one type) or bimodal (two type) networks.45 Furthermore, despite being built for unimodal data, if a user asks Gephi to run network algorithms on bimodal data, Gephi continues to use unimodal formulas and ‘everything still works as though these metrics would produce meaningful, sensible results’.46

这个平台的目标，是让用户能够生成自己的结果和可视化；它通过把检索结果直接显示为包括网络图在内的视觉形式来鼓励这种主动探索。对近代早期爱尔兰感兴趣的用户，因此可以搜索并可视化Katherine Philips（1632 – 64）、Anne Southwell（1574? – 1636）和Dorothy Moore（1613 – 64）等女性的接受网络。以Katherine Philips为例，用户能够直观看出是谁在编纂她的作品、她的作品被编入哪些类型的材料，以及谁在写到她。和大多数人文学科数据一样，RECIRC生成的网络十分复杂，全部属于双模或多模网络，也就是说，其中包含两种或更多类型的节点。在这些网络中，节点（行动者）可能属于四种类型之一：女性作者、接受者、所有者/编纂者/抄写者，以及接受来源类型。如果某一网络只包含四种类型中的两种，例如女性作者和接受者，那么它就是双模网络；如果包含三种或更多类型，则属于多模网络。Scott Weingart指出，人文学科数据经常包含丰富多样的节点类型，例如人物、地点和事物，但“你的数据集越复杂、越多模，你能够合理完成的分析反而越少”。44 他还指出，在双模网络中，“度中心性不过是对归属关系的计数”，而且“聚类系数最简单的定义并不适用于双模网络”。对于只具有网络分析初级知识的用户来说，这些问题又会被开源软件本身的局限进一步放大。例如Gephi无法区分单模（一种节点类型）与双模（两种节点类型）网络。45 更重要的是，虽然Gephi的算法是为单模数据设计的，但如果用户让它直接对双模数据运行网络算法，软件仍会继续套用单模公式，而且“所有东西看起来仍然正常工作，就好像这些指标真的能够产生有意义、合理的结果”。46

To avoid this issue, the approach taken by RECIRC’s network visualisations is qualitative in nature and is exemplified by the work of Bronagh Ann McShane.47 She draws on a method Van Vugt describes as ‘disclose’ (a mixture between close and distant reading), and her dataset is drawn from the RECIRC database. McShane’s analysis is on data from 405 letters containing reception evidence and written between 1609 and 1693. This data, was gathered from the Archive of the Archdiocese of

Mechelen and relates to letters from Lady Mary Percy and other nuns from the English Benedictine Convent of the Assumption of Our Blessed Lady to the archbishop of Mechelen and his secretaries. McShane's analysis of these receptions does not employ structural algorithms, except degree measures, which she uses to count affiliations and bring attention to both female authors and agents of reception with large amounts of edges. Instead, she argues that filtering the network allows for a more nuanced understanding of the types of reception that takes place. She achieves this by highlighting how filtering the network to 'identify receptions that are translations only' reveals alliances between the nuns and their translators that are more complex than scholars previously thought.48 McShane's case study points to ways in which RECIRC's broader research questions can be visualised and analysed using a qualitative network approach. Degree measurements such as in-degree and out-degree can be used by users to ask what female authors were read/written about, and who was reading/writing about female authors. Through the use of filters, users can engage with network visualisations to explore the different types of reception that occurred, and by joining McShane's approach with Van Vugt's multi-layering approach, close reading is facilitated as users can explore the types of sources a reception takes place in and even interact with the network in a way that will bring them to a transcript of the reception evidence. Overall, RECIRC's platform highlights the potential for interactive qualitative networks as a way to facilitate close reading of the reception of women authors, while also enabling more experienced users to potentially grapple with the multimodal networks. Those scholars with the technical capabilities could condense the networks in order to ask new research questions of the data: for example, which female authors were compiled together in miscellanies or auction catalogues; and which types of reception frequently occur together (for example, how many translations also contain a reference to a specific work and/or female author)?

为避免这一问题，RECIRC的网络可视化采取了以定性解释为主的方法，Bronagh Ann McShane的研究就是典型案例。47 她采用Van Vugt所称的“ disclose ”方法，也就是近读与远读的结合，数据则来自RECIRC数据库。McShane分析的是405封包含接受证据、写于1609至1693年之间的书信。这些数据采自Mechelen总教区档案，涉及Lady Mary Percy以及“ 圣母升天英格兰本笃会修道院 ”其他修女写给Mechelen大主教及其秘书的书信。McShane分析这些接受关系时，除了度数测量之外并不使用结构性算法；度数只是用来计数归属关系，并把注意力引向拥有大量连接的女性作者和接受传播代理人。她真正强调的是，通过筛选网络，可以更细致地理解不同类型的接受。她举例指出，把网络筛选到“ 只识别属于翻译的接受 ”时，会显示出修女与翻译者之间比以往学者所认识到的更加复杂的联盟。48 McShane的个案说明，RECIRC更广泛的研究问题可以如何通过定性网络方法进行可视化和分析。用户可以利用入度、出度等度数指标，追问哪些女性作者被阅读或被写到，以及是谁在阅读或书写这些女性作者。通过筛选器，用户能够借助网络可视化探索不同的接受类型；如果把McShane的方法与Van Vugt的多层网络方法结合，研究者还能够展开近读：不仅查看某一次接受发生于什么类型的来源中，甚至可以与网络互动，直接进入该接受证据的文字转录。总体来看，RECIRC平台显示了互动式定性网络在促进对女性作者接受史进行细读方面的潜力，同时也使更有经验的用户有可能继续处理多模网络。具备较强技术能力的研究者，还可以对网络进行压缩，以便从数据中提出新问题，例如：哪些女性作者经常被一起收入杂抄本或拍卖目录？哪些接受类型经常同时出现，例如有多少翻译文本又同时提到某一部具体作品和/或某位女性作者？

IV

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While the work of TNOP, SDFB, MACMORRIS and RECIRC has shown how the approach offers many opportunities for those interested in the recovery of women's lives, it also brings with it many challenges, both for the individual researcher and for the community as a whole. The first issue we face is surmounting the technical learning curve. One of the most pertinent issues in this regard is the black-box effect — 'inputting data and getting results without fully understanding the algorithms through which those results were generated'.49 As noted, software like Gephi is unable to differentiate between unimodal (one type) or bimodal (two type) networks. Thus, if a scholar did not know if they were working with unimodal or bimodal data, they would not know that the results being generated might not be reliable. Ahnert, Ahnert, Coleman and Weingart argue that 'use of any of these tools requires a prior mental manoeuvre of translating cultural artefacts into an abstracted

form to see whether they are compatible with the input requirements of the available tools'.50 This involves taking the time to engage with background reading; gaining familiarity with platforms like Gephi, Palladio or Nodegoat through tutorials; experimenting with test data; and experimenting with different ways of structuring data. It may also lead to a need or desire to develop skills further, such as learning how to code or learning how design your own database. This can pose a challenge for potential digital scholars as it can be difficult to tell at an early stage whether the methods offered by network analysis will prove fruitful, which can be daunting for people who have a limited timeframe to complete their projects. However, the benefits of being digitally engaged in a rapidly evolving digital world will for some outweigh the challenges.

TNOP、SDFB、MACMORRIS和RECIRC已经表明，网络方法为希望恢复女性生活的研究者提供了许多机会，但它同时也带来不少挑战，既影响个体研究者，也影响整个学术共同体。首先需要克服的是技术学习曲线。其中最突出的问题之一，是“黑箱效应”——“把数据输入进去，再得到结果，却并不真正理解这些结果是通过什么算法生成的”。49 如前所述，Gephi之类的软件无法区分单模网络与双模网络。因此，如果研究者根本不知道自己处理的是单模还是双模数据，就不会意识到软件生成的结果可能并不可靠。Ahnert、Ahnert、Coleman和Weingart指出，“使用任何这类工具，都要求研究者先在思维上完成一个动作：把文化材料转译成抽象形式，再判断它是否符合现有工具的输入要求”。50 这意味着研究者需要花时间阅读背景材料，通过教程熟悉Gephi、Palladio或Nodegoat等平台，用测试数据反复实验，并尝试不同的数据结构方式。这还可能进一步带来学习编程、设计数据库等技能的需要或愿望。对潜在的数字研究者来说，这会构成挑战，因为在研究早期很难判断网络分析提供的方法究竟是否会产生有价值的结果；而对于项目期限有限的人，这种不确定性尤其令人却步。不过，对一些研究者而言，在迅速数字化的世界中积极参与数字研究所带来的收益，仍然可能大于这些挑战。

This highlights a significant and pervasive issue: autodidactism, or the self-taught person, is a major feature of digital humanities in Ireland and the reasons for this are both logistical and structural. Logistically, as noted by Coolahan, 'it is hard to recruit computer science specialists in today's job market, where the financial rewards are far greater in industry than the humanities'.51 However, there is also a structural element to this because we are in a moment where research funding bids are more successful if they include a digital component. In the case of a researcher applying for individual funding, this may require the autodidactic acquisition of new skills, but for large research projects, the issues are slightly different. In the absence of computer science specialists, the positions on these projects need to be filled by people who have both domain knowledge of the project's research area and the digital skills needed to achieve the project's aims. People who fulfil both of these criteria are a rare breed, so projects must ensure they hire those with the best domain knowledge and potential to gain the relevant digital skillset. This 'upskilling' often occurs through autodidactic means, with the provision of some formal support. This is the route that I took, but it is a privileged one. When joining RECIRC as a Ph.D. researcher I had a background in early modern women's writing and no digital training. As part of the project, I had access to funding, a support network, the space to watch how the project was operating and the time to experiment with different digital methods. I was then fortunate to be able to formalise this training through a training school (funded by the Arts and Humanities Research Council) in the U.K. This ad hoc pathway is not sustainable. Considering the U.K. context, Ahnert, Ahnert, Coleman and Weingart contend that 'a pressing duty is placed on the university to prepare future generations of academics by offering suitable combinations of courses in humanities subjects, programming, and statistical methods'.52 The same is needed in Ireland, and we need to make sure that it can thrive, not just survive. James O'Sullivan has noted that, 'DH programmes, initiatives and cohorts have had a tendency to implode over the years, suggesting that individuals rather than institutions are driving the digital humanities in Ireland'.53 In order to avoid this continuing, we need to work together as a community and draw on the experience of the institutions and practitioners that have begun to have some success in implementing digital humanities programmes. Most of the projects discussed in this paper had access to some form of digitised material — digital surrogates of primary sources, metadata shared from other digitisation projects or machine-readable text sourced from biographical

这进一步暴露出一个重要而普遍的问题：自学（autodidactism）是爱尔兰数字人文学的重要特征，其原因既有现实层面的，也有结构层面的。现实上，正如Coolahan指出的，“在今天的就业市场上很难招到计算机科学专家，因为产业界的经济回报远高于人文学科”。51 但问题也具有结构性，因为在目前的环境中，研究经费申请如果包

含数字组成部分，往往更容易成功。对于申请个人资助的研究者来说，这可能意味着需要靠自学获得新技能；对于大型研究项目，问题则略有不同。如果缺少计算机科学专家，项目岗位就必须由那些既熟悉研究领域本身、又具备实现项目目标所需数字技能的人来承担。两种条件都完全具备的人非常少，因此项目必须确保雇用那些领域知识最扎实、同时最有潜力获得相关数字技能的人。这种“技能提升”往往主要依靠自学，同时辅以一定形式的正式支持。我自己的经历就是如此，但这其实是一条带有特权性质的路径。加入RECIRC担任博士研究员时，我有近代早期女性写作的研究背景，却没有任何数字训练。项目为我提供了经费、支持网络、观察项目如何运作的空间，以及尝试不同数字方法的时间。后来我又很幸运地能够参加英国一个由Arts and Humanities Research Council资助的培训学校，把这些零散训练正式化。但这种临时拼接出来的培养路径并不可持续。就英国情况而言，Ahnert、Ahnert、Coleman和Weingart认为，大学“承担着一项紧迫责任：通过提供人文学科、编程和统计方法的适当课程组合，为未来一代学者做好准备”。52

爱尔兰也需要这样做，而且必须确保数字人文能够真正发展，而不是仅仅维持生存。James O’Sullivan指出，“多年来，数字人文项目、倡议和研究群体往往有自行瓦解的倾向，这说明爱尔兰的数字人文更多是由个人而不是机构推动的”。53 如果不希望这种情况继续，就需要整个共同体协作，并借鉴那些已经在建设数字人文项目方面取得一定成效的机构与实践者的经验。本文讨论的大多数项目都能够获得某种形式的数字化材料——原始史料的数字复制品、其他数字化项目共享的元数据，或者来自传记资料的机器可读文本。

authorities. Those that did not undertook large data-collection phases, which involved coordinated transcription efforts in conjunction with the recording of metadata in line with a predetermined taxonomy. In terms of material of interest to scholars of early modern women’s lives, it is more likely that the material is not available in a digital form and is likely scattered across several archives. This means that interested scholars would need to go down the second route and the effort needed to find, gather, transcribe and pre-process the data for analysis is exponentially higher, and thus the research might not be feasible without some form of research funding. There are significant challenges with the material that has been digitised or already has metadata. As an extreme example, let us examine the state papers. RECIRC has shown that reading the content of letters for receptions of women writers can reveal new information about early modern women’s lives and that receptions of women can far outweigh the number of extant letters to or from a particular woman. However, TNOP has focused exclusively on the senders and recipients of letters and when you consider they analysed 132,747 letters, it is not hard to see why. If a scholar was to attempt this daunting task of reading the state papers for reception evidence of women’s lives, it is likely that they would have to run digitised facsimiles of all the letters through a programme like Transkribus in order to transform the manuscript image into machine readable text. They would then have to correct any OCR errors before running the documents through NER to extract names from the documents. What would follow would be an extensive data cleaning phase aimed at deduplicating and disambiguating the various people extracted while also deciding on how they categorise those mentioned within the letters. The final step would be to connect the people mentioned in the letter to the writer to create a reception network, or connect the people mentioned to each other to create a co-citation network. This mammoth task could take years, even if you only examined a small portion of the letters. The alternative would be to read through the letters with a specific question in mind, marking up the mentions in a way that allows you to answer this question, but this would only be possible with a much smaller dataset. This means that a big task for feminist scholars interested in the potential of data-driven networked approaches is the ability to create feasible projects when confronted with the gendered realities of how metadata has been created to date and how/what has been thus far digitised. One final challenge I wish to highlight was succinctly addressed in a question posed by Dorothea Salo: ‘is there a readable, reviewable, (print-) publishable, citable, immutable, preservable text in these data?’54 This highly pertinent question raises the need for discussion on issues of sustainability, what work we as humanities scholars privilege and how that work is presented. These are questions too large to grapple with in this piece, but I do wish to discuss the impact that the privileging of print can have on networked approaches. Salo argues that the link between print publication as a marker of humanities career progress can pose a challenge for digital humanists as ‘their nontextual research products are usually not printable, if printable at all, without loss of function’.55 In terms of the projects

没有这类现成材料的项目，则必须经历规模很大的数据收集阶段：组织协调转录工作，同时按照预先设定的分类体系记录元数据。对于研究近代早期女性生活的学者来说，真正感兴趣的材料更可能尚未数字化，而且分散在多个档案馆中。这意味着研究者只能采取后一条路径；寻找、汇集、转录并为分析预处理数据所需要的劳动会成倍增加，因此如果没有某种研究经费支持，项目可能根本不可行。即使材料已经数字化或已经存在元数据，也仍有很大挑战。以国家档案为一个极端例子：RECIRC已经证明，通过阅读书信正文，寻找女性作者的“ 接受 ”证据，可以发现关于近代早期女性生活的新信息；而一位女性被他人提到的接受证据数量，可能远远超过现存写给她或由她写出的书信数量。TNOP却只分析书信的寄件人与收件人，考虑到他们处理了132,747封信，这个选择并不难理解。如果某位学者真的想承担这项艰巨任务，通读国家档案以寻找关于女性生活的接受证据，他很可能首先需要把所有书信的数字化影像送入Transkribus一类程序，将手稿图像转换成机器可读文本；随后纠正OCR错误，再用NER从文本中抽取人名。接下来还要经历一轮大规模数据清洗：去除重复人物、消除同名异人或异名同人的歧义，并决定如何给信中提到的人物分类。最后一步，是把信中提到的人与写信者连接起来，建立接受网络；或者把信中被提到的人彼此连接，建立共被引网络。即使只考察其中很小一部分书信，这样的巨大工程也可能需要数年时间。另一种选择，是带着一个非常具体的问题阅读书信，只标记与该问题有关的人物提及；但这种方式只能处理小得多的数据集。因此，对于希望利用数据驱动网络方法的女性主义学者来说，一个重大任务，就是面对既有元数据如何被生产、以及哪些材料迄今被数字化这些带有性别结构的现实，仍然设计出可行的研究项目。我还想强调最后一个挑战。Dorothea Salo用一个问题简洁地提出了它：“ 这些数据中，是否存在一种可阅读、可评审、可（印刷）出版、可引用、不可篡改、可保存的文本？ ”54 这个问题非常关键，因为它迫使我们讨论可持续性、人文学者究竟优先承认哪些工作，以及这些工作应如何呈现。本文无法完整处理这些过于庞大的问题，但我想特别讨论“ 印刷优先 ”如何影响网络研究。Salo指出，把印刷出版视为人文学术职业进步的标志，会给数字人文学者造成困难，因为“ 他们的非文本研究成果通常无法打印；即使能够打印，也会失去功能 ”。55
对于本文讨论的项目而言，

discussed in this piece, this includes the inter-operable network graphs developed by SDFB, TNOP and RECIRC. The graphs themselves need to be given equitable recognition when compared to the traditional outputs by these projects, as it is these graphs that can reveal things never seen before about the lives of early modern women. As we also privilege what we can see, we can very easily overlook all the collection, cleaning, design and development work that goes into creating these graphs. Thus, the challenge is to move beyond the single author model prevalent in the humanities and ensure that all members linked to a project (such as principal investigators, research assistants, postdocs, research technologists and Ph.D. researchers) are credited for the work they do and that all forms of labour are no longer invisible, in that they are properly valued and equally considered when it comes to career progression. While these are all huge challenges, they are not unique to the study of women in early modern Ireland, nor even to early modern studies in general. However, at the ten-year celebration of the launch of the 1641 depositions Jane Ohlmeyer noted that feminist scholars were one of the first to use the digitised version of depositions in their scholarship, as seen by the work of McAreavey, Coolahan and Joan Redmond.56 If you also consider the ground-breaking work achieved by Perdita, RECIRC and the Pulter Project, it is clear that there is a close connection between the study of women in early modern Ireland and digital humanities. Since the publication of the original ‘Agenda’, advancements in digital technology have meant that the study of Irish women’s history can take new directions that were never envisaged by MacCurtain, O’Dowd and Luddy. These new directions can draw on the success of Perdita, RECIRC and the Pulter Project and can engage with new techniques such as network visualisation and analysis. However, if we want to continue with the tradition started by these projects, we cannot ignore the wider challenges, nor can we be passive observers of these debates. As Catherine D’Ignazio and Lauren Klien have argued, feminism is about power and in our contemporary world, data is power.57 As feminist scholars, we need actively to engage with these challenges.58

这包括SDFB、TNOP和RECIRC开发的可互操作网络图。与这些项目的传统成果相比，网络图本身也应当获得同等认可，因为正是这些图能够揭示此前从未被看见的近代早期女性生活面貌。与此同时，由于我们也倾向于优先重视“ 看得见 ”的成果，就很容易忽略生成这些图背后全部的数据收集、清洗、设计和开发劳动。因此，一个重要挑战是走出人文学科长期占主导地位的“ 单一作者 ”模式，确保所有与项目相关的成员——例如首席研究员、研究助理、博士后、研究技术人员和博士研究员——都能够为自己的工作获得署名，也确保各种形式的劳动不再隐形，而是在职业发展中得到适当评价并被平等考虑。虽然这些都是巨大挑战，但它们并不只属于近代早期爱尔兰

女性研究，甚至也不只属于近代早期研究。可是，在1641年证词数字项目上线十周年纪念活动上，Jane Ohlmeyer指出，女性主义学者是最早把数字化证词用于研究的群体之一，这一点可以从McAreavey、Coolahan和Joan Redmond的工作中看到。56 如果再考虑Perdita、RECIRC和Pulter Project所取得的开创性成果，就会发现近代早期爱尔兰女性研究与数字人文之间存在非常紧密的联系。自最初的《研究议程》发表以来，数字技术的进步已经使爱尔兰女性史能够走向MacCurtain、O ’ Dowd和Luddy当年无法预见的新方向。这些新方向可以继承Perdita、RECIRC和Pulter Project的成功，也可以采用网络可视化和网络分析等新技术。不过，如果希望延续这些项目开启的传统，我们就不能忽视更广泛的挑战，也不能只是被动旁观这些争论。正如Catherine D ’ Ignazio和Lauren Klein所说，女性主义关乎权力，而在当代世界中，数据就是权力。57
作为女性主义学者，我们必须主动介入并处理这些挑战。58

Notes

注释

参考文献类注释中的作者名、题名、出版信息与网址保留原文，以便准确检索；说明性文字译为中文。

* Department of English, Maynooth University, evan.bourke@mu.ie

* 梅努斯大学英语系，evan.bourke@mu.ie

1. Mark E. Newman, Networks: an introduction (Oxford, 2010).

1. 文献条目（作者名、题名、出版信息与网址保留原文以便检索）：Mark E. Newman, Networks: an introduction (Oxford, 2010).

2. See also, Catherine Medici, 'Using network analysis to understand early modern women' in Early Modern Women: An Interdisciplinary Journal, xiii, no.1 (Fall 2018), p. 155.

2. 另见：Catherine Medici, ‘ Using network analysis to understand early modern women ’ in Early Modern Women: An Interdisciplinary Journal, xiii, no.1 (Fall 2018), p. 155。

3. See Newman, Networks and Albert-László Barabási, Linked: the new science of networks (Cambridge, MA, 2002).

3. 文献条目（作者名、题名、出版信息与网址保留原文以便检索）：See Newman, Networks and Albert-L á szl ó Barab á si, Linked: the new science of networks (Cambridge, MA, 2002).

4. See, for example, Jenna Townend, 'Quantitative and qualitative approaches to early-modern networks: the case of George Herbert (1593–1633) and his imitators' in Literature Compass, xiv, no. 3 (Mar. 2017), pp 1–14; Anupam Basu, Jonathan Hope and Michael Witmore, 'The professional and linguistic communities of early modern dramatists' in Anthony W. Johnson, Roger D. Sell and Helen Wilcox (eds), Community-making in early Stuart theatres: stage and audience (London, 2016), pp 63–94.

4. 例如，参见：Jenna Townend, ‘ Quantitative and qualitative approaches to early-modern networks: the case of George Herbert (1593 – 1633) and his imitators ’ in Literature Compass, xiv, no. 3 (Mar. 2017), pp 1 – 14；Anupam Basu, Jonathan Hope and Michael Witmore, ‘ The professional and linguistic communities of early modern dramatists ’ in Anthony W. Johnson, Roger D. Sell and Helen Wilcox (eds), Community-making in early Stuart theatres: stage and audience (London, 2016), pp 63 – 94。

5. Ruth Ahnert, Sebastian E. Ahnert, Catherine Nicole Coleman and Scott B. Weingart, The network turn: changing perspectives in the humanities (Cambridge, 2021), pp 5–7.

5. 文献条目（作者名、题名、出版信息与网址保留原文以便检索）：Ruth Ahnert, Sebastian E. Ahnert, Catherine Nicole Coleman and Scott B. Weingart, The network turn: changing perspectives in the humanities (Cambridge, 2021), pp 5 – 7.

6. Medici, 'Using network analysis', p. 153.

6. 文献条目（作者名、题名、出版信息与网址保留原文以便检索）：Medici, ‘ Using network analysis ’, p. 153.

7. Medici, 'Using network analysis', pp 153–4; Susan Broomhall and Stephanie Tarbin (eds), Women, identities, and communities in early modern Europe (Burlington, 2008); Bernard Capp, When gossips meet: women, family, and neighbourhood in early modern England (Oxford, 2004); Amanda E. Herbert, Female alliances: gender, identity, and friendship in early modern Britain (New Haven, 2014); James Daybell, Women letter writers in Tudor England (Oxford, 2006); Julie Crawford, Mediatrix: women, politics, and literary production, in early modern England (Oxford, 2014); Micheline White, English women, religion, and textual production 1500–1625 (Burlington, 2011); Julie Campbell, Literary circles and gender in early modern Europe: a cross cultural approach (Burlington, 2007).

7. 文献条目（作者名、题名、出版信息与网址保留原文以便检索）：Medici, ‘ Using network analysis ’, pp 153 – 4; Susan Broomhall and Stephanie Tarbin (eds), Women, identities, and communities in early modern Europe (Burlington, 2008); Bernard Capp, When gossips meet: women, family, and neighbourhood in early modern England (Oxford, 2004); Amanda E. Herbert, Female alliances: gender, identity, and friendship in early modern Britain (New Haven, 2014); James Daybell, Women letter writers in Tudor England (Oxford, 2006); Julie Crawford, Mediatrix: women, politics, and literary production, in early modern England (Oxford, 2014); Micheline White, English women, religion, and textual production 1500 – 1625 (Burlington, 2011); Julie Campbell, Literary circles and gender in early modern Europe: a cross cultural approach (Burlington, 2007).

8. Margaret MacCurtain and Mary O'Dowd (eds), Women in early modern Ireland (Edinburgh, 1991).

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9. Margaret MacCurtain, Mary O'Dowd and Maria Luddy, 'An agenda for women's history in Ireland, 1500–1900' in I.H.S., xxviii, no. 109 (May 1992), pp 6, 8, 11.

9. 文献条目（作者名、题名、出版信息与网址保留原文以便检索）：Margaret MacCurtain, Mary O ’ Dowd and Maria Luddy, ‘ An agenda for women ’ s history in Ireland, 1500 – 1900 ’ in I.H.S., xxviii, no. 109 (May 1992), pp 6, 8, 11.

10. For a list of work on early modern Irish women since the publication of the agenda, see the appendix in Julie A. Eckerle and Naomi McAreavey (eds), Women's life writing & early modern Ireland (Lincoln, NE, 2019), pp 253–80.

10. 有关《研究议程》发表后近代早期爱尔兰女性研究的文献清单，见 Julie A. Eckerle and Naomi McAreavey (eds), Women ’ s life writing & early modern Ireland (Lincoln, NE, 2019) 的附录，pp 253 – 80。

11. Jerrold Casway, ‘Heroines or victims? The women of the flight of the earls’ in New Hibernia Review, vii, no. 1 (spring 2003), pp 57–74; idem, ‘Rosa O Dogherty: a Gaelic woman’ in Seanchas Ardmhacha: Journal of the Armagh Diocesan Historical Society, x, no. 1 (1980–81), pp 42–62; Marie-Louise Coolahan, ‘Irish women’s letters, 1641–1653’ in James Daybell and Andrew Gordon (eds), Women and epistolary agency in early modern culture (Abington, 2016), pp 167–81; Karen Ann Holland, ‘Joan Desmond, Ormond and Ossory: the world of a countess in sixteenth-century Ireland’ (Ph.D. thesis, Providence College, 1995).

11. 文献条目（作者名、题名、出版信息与网址保留原文以便检索）：Jerrold Casway,‘ Heroines or victims? The women of the flight of the earls ’ in New Hibernia Review, vii, no. 1 (spring 2003), pp 57 – 74; idem,‘ Rosa O Dogherty: a Gaelic woman ’ in Seanchas Ardmhacha: Journal of the Armagh Diocesan Historical Society, x, no. 1 (1980 – 81), pp 42 – 62; Marie-Louise Coolahan,‘ Irish women ’ s letters, 1641 – 1653 ’ in James Daybell and Andrew Gordon (eds), Women and epistolary agency in early modern culture (Abington, 2016), pp 167 – 81; Karen Ann Holland,‘ Joan Desmond, Ormond and Ossory: the world of a countess in sixteenth-century Ireland ’ (Ph.D. thesis, Providence College, 1995).

12. Deirdre Fennell, ‘The reluctant lord deputy: the early life and career of Sir William Fitzwilliam of Milton’ (Ph.D. thesis, N.U.I. Galway, 2018); eadem, ‘Female presence in the life of Lord Deputy Sir William Fitzwilliam’, Tudor and Stuart Ireland Podcasts 2016 (https://soundcloud.com/history-hub/deirdre-fennell-female-presence-in-life-of-lord-deputysir-william-fitzwilliam?in=history-hub/sets/tudor-and-stuart-ireland-2016); Catherine Medici, ‘To persuade and connect: Mary Sidney’s essential role in Henry Sidney’s Irish rule’ in Laura Aydelotte (ed.), A mirror for medieval and early modern studies: selected proceedings of the Newberry Center for Renaissance Studies 2012 Multidisciplinary Graduate Student Conference (Chicago, 2012), pp 61–72.

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13. Coolahan, ‘Irish women’s letters, 1641–1653’, pp 167–81; Eleanor O’Keeffe, ‘The family and marriage strategies of James Butler, first Duke of Ormonde, 1658–1688’ (Ph.D. thesis, University of Cambridge, 2000); Naomi McAreavey, ‘The place of Ireland in the letters of the first Duchess of Ormonde’ in Eckerle and McAreavey (eds), Women’s life writing, pp 159–182; Damien Duffy, Aristocratic women in Ireland, 1450–1660: the Ormond family, power and politics (Woodbridge, 2021); Ruth Connolly, ‘“A wise and godly Sybilla”: Viscountess Ranelagh and the politics of international Protestantism’ in Sylvia Brown (ed.), Women, gender and radical religion in early modern Europe (Leiden, 2007), pp 285–306; eadem ‘A proselytising Protestant commonwealth: the religious and political ideals of Katherine Jones, Viscountess Ranelagh’ in The Seventeenth Century, xxiii, no. 2 (2008), pp 244–64; Michelle DiMeo, ‘The rhetoric of medical authority in Lady Katherine Ranelagh’s letters’ in Daybell and Gordon (eds), Women and epistolary agency, pp 96–109; eadem ‘“Such a sister became such a brother”: Lady Ranelagh’s influence on Robert Boyle’ in Intellectual History Review, xxv, no. 1 (2015), pp 21–36; Clodagh Tait, ‘Good ladies and ill wives: women on Boyle’s estates’ in David Edwards and Colin Rynne (eds), The colonial world of Richard Boyle first earl of Cork, (Dublin, 2018), pp 205–22; Betsey Taylor-Fitzsimon, ‘Conversion, the Bible and the Irish language: the correspondence of Lady Ranelagh and Bishop Dopping’ in Michael Brown, Charles McGrath and Thomas Power (eds), Converts and conversion in Ireland, 1650–1850 (Dublin, 2005), pp 157–82; Carol Pal, Republic of women: rethinking the republic of letters in the seventeenth century (Cambridge, 2012); Ann-Maria Walsh, The daughters of the first Earl of Cork: writing family, faith, politics and place (Dublin, 2020); Ramona Wray, ‘[Re]constructing the past: the diametric lives of Mary Rich’ in Henk Dragsta, Sheila Ottway, and Helen Wilcox (eds), Betraying our selves: forms of self-representation in early modern English texts (London, 2000), pp 148–65; Amelia Zurcher, ‘Life writing in the Boyle family network’ in Eckerle & McAreavey (eds), Women’s life writing, pp 99–136.

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14. The main exception is Medici and my work, discussed below.

14. 主要的例外是 Medici 的研究和我本人在下文讨论的研究。

15. RECIRC (The Reception and Circulation of Early Modern Women's Writing, 1550–1700) (recirc.nuigalway.ie) (29 Oct. 2020).

15. 文献条目（作者名、题名、出版信息与网址保留原文以便检索）：RECIRC (The Reception and Circulation of Early Modern Women ' s Writing, 1550 – 1700) (recirc.nuigalway.ie) (29 Oct. 2020).

16. For more on this, see Emilie K. M. Murphy, 'Exile and linguistic encounter: early modern English convents in the low countries and France' in Renaissance Quarterly, lxxiii, no. 1 (spring 2020), pp 132–64; eadem, 'Language and power in an English convent in exile, c.1621–c.1631' in Hist. Jn., lxii, no. 1 (Mar. 2019), pp 101–25; Bronagh Ann McShane, 'Visualising the reception and circulation of early modern nuns' letters' in Journal of Historical Network Research, ii (2018), pp 1–25.

16. 关于这一问题的更多讨论，见：Emilie K. M. Murphy, ' Exile and linguistic encounter: early modern English convents in the low countries and France ' in Renaissance Quarterly, lxxiii, no. 1 (spring 2020), pp 132 – 64；eadem, ' Language and power in an English convent in exile, c.1621 – c.1631 ' in Hist. Jn., lxii, no. 1 (Mar. 2019), pp 101 – 25；Bronagh Ann McShane, ' Visualising the reception and circulation of early modern nuns ' letters ' in Journal of Historical Network Research, ii (2018), pp 1 – 25。

17. Stephen Borgatti, 'Centrality and network flow' in Social Networks, xxvii, no. 1 (Jan. 2005), pp 55–71.

17. 文献条目（作者名、题名、出版信息与网址保留原文以便检索）：Stephen Borgatti, ' Centrality and network flow ' in Social Networks, xxvii, no. 1 (Jan. 2005), pp 55 – 71.

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18. 文献条目（作者名、题名、出版信息与网址保留原文以便检索）：Evan Bourke, ' Female involvement, membership and centrality: a social network analysis of the Hartlib Circle ' in Literature Compass, xiv, no. 4 (Apr. 2017), pp 1 – 17.

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22. Ahnert & Ahnert, ' Protestant letter networks ', pp 3 – 4。另见 Ingeborg Van Vugt, ' Using multi-layered networks to disclose books in the republic of letters ' in Journal of Historical Network Research, i (2017), pp
25 – 51；该文强调书信网络，并考察书籍共享在强化通信关系中的作用。

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Empire。该项目从1610至1833年葡萄牙帝国的170,000封行政通信中提取网络数据：Agata B ł och, Dernival Vasques Filho and Michael Bojanowski, ' Networks from archives: reconstructing networks of official correspondence in the early modern Portuguese empire ' in Social Networks, lxix (May 2022), pp 123 – 35。

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27. https://catherinemediciphd.org/visualizing-the-sidney-network/; The Hartlib Papers Online (https://www.dhi.ac.uk/hartlib/) (11 Jan. 2021).

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29. 文献条目（作者名、题名、出版信息与网址保留原文以便检索）：Ahnert & Ahnert,‘ Protestant letter networks ’, pp 3 – 4; McShane,‘ Visualising the reception ’, pp 1 – 2.

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35. Marie-Louise Coolahan, 'Dowdall (née Southwell), Elizabeth (fl. 1640–1642)', Ioanna Tsakiropoulou, 'Devereux, Frances [née Walsingham; other married names Sidney and Burke] (c.1568–1632)' in O.D.N.B. (https://www.oxforddnb.com).

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36. For an alternative overview of these projects, see Marie-Louise Coolahan, 'New technologies of research and digital interpretation for early modern Irish Studies' in Irish University Review, l, no. 1 (May 2020), pp 175–86.

36. 关于这些项目的另一种概述，见 Marie-Louise Coolahan,‘ New technologies of research and digital interpretation for early modern Irish Studies ’ in Irish University Review, l, no. 1 (May 2020), pp 175 – 86。

37. David Baker, Willy Maley and Patricia Palmer, 'What ish my network? Introducing MACMORRIS: digitising cultural activity and collaborative networks in early modern Ireland' in Literature Compass, xv, no. 11 (Nov. 2018), p. 4. See also Patricia Palmer, David J. Baker, and Willy Maley, 'Enter MacMorris' in Dublin Review of Books, 114 (July 2019) (https://www.drb.ie/essays/enter-macmorris) (27 June 2022).

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38. We are thankful to the Royal Irish Academy and especially Turlough O'Riordan for sharing the D.I.B. metadata with us, and to Katharine Simms, Michael Hoyne and the Dublin Institute of Advanced Studies for sharing the BPD with us.

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39. Warren et al., 'Six degrees of Francis Bacon', p. 4.

39. 文献条目（作者名、题名、出版信息与网址保留原文以便检索）：Warren et al.,‘ Six degrees of Francis Bacon ’, p. 4.

40. Bernadette Cunningham, 'Women and Gaelic literature, 1500–1800' in MacCurtain & O'Dowd (eds), Women in early modern Ireland, pp 147–8.

40. 文献条目（作者名、题名、出版信息与网址保留原文以便检索）：Bernadette Cunningham,‘ Women and Gaelic literature, 1500 – 1800 ’ in MacCurtain & O ' Dowd (eds), Women in early modern Ireland, pp 147 – 8.

41. For more on women mentioned within the poems, see Damian McManus, 'Celebrating the female in Classical Irish poetry: the wife' in Ériu, 65 (2015), pp 137–68; idem, 'Female ancestry and mother's kin in Classical Irish poetry' in Caoimhín Breatnach and Meidhbhín Ní Úrdail (eds), Aon don Éigse. Essays marking Osborn Bergin's centenary lecture on Bardic poetry (1912) (Dublin, 2015), pp 193–219.

41. 关于诗歌中提到的女性，更多参见 Damian McManus,‘ Celebrating the female in Classical Irish poetry: the wife ’ in É riu, 65 (2015), pp 137 – 68；idem,‘ Female ancestry and mother ’ s kin in Classical Irish poetry ’ in Caoimh í n Breatnach and Meidhbh í n N í Ú rdail (eds), Aon don É igse. Essays marking Osborn Bergin ’ s centenary lecture on Bardic poetry (1912) (Dublin, 2015), pp 193 – 219。

42. Coolahan, 'New technologies of research', p. 182.

42. 文献条目（作者名、题名、出版信息与网址保留原文以便检索）：Coolahan,‘ New technologies of research ’, p. 182.

43. Ibid.

43. 同上。

44. Scott Weingart, 'Networks demystified 9: bimodal networks', The Scottbot Irregular, 21 Jan. 2015 (https://scottbot.net/networks-demystified-9-modality/) (11 Jan. 2021).

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45. https://gephi.org/ (11 Jan. 2021).

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46. Ibid. This is especially an issue with betweenness centrality and the creators of Gephi are aware of this problem.

46. 同上。这个问题在中介中心性方面尤其突出，Gephi 的开发者也意识到了这一问题。

47. McShane, 'Visualising the reception', pp 1–25.

47. 文献条目（作者名、题名、出版信息与网址保留原文以便检索）：McShane,‘ Visualising the reception ’, pp 1 – 25.

48. Ibid., p. 15.

48. 同上，p. 15。

49. Ahnert et al., The network turn, p. 85.

49. 文献条目（作者名、题名、出版信息与网址保留原文以便检索）：Ahnert et al., The network turn, p. 85.

50. Ibid., p. 75.

50. 同上，p. 75。

51. Coolahan, 'New technologies of research', p. 184.

51. 文献条目（作者名、题名、出版信息与网址保留原文以便检索）：Coolahan,‘ New technologies of research ’, p. 184.

52. Ahnert et al., The network turn, p. 86.

52. 文献条目（作者名、题名、出版信息与网址保留原文以便检索）：Ahnert et al., The network turn, p. 86.

53. James O'Sullivan, 'The digital humanities in Ireland' in Digital Studies/Le Champ Numérique, x, no. 1 (2020), p. 23.

53. 文献条目（作者名、题名、出版信息与网址保留原文以便检索）：James O ' Sullivan,‘ The digital humanities in Ireland ’ in Digital Studies/Le Champ Num é rique, x, no. 1 (2020), p. 23.

54. Dorothea Salo, 'Is there a text in these data? The digital humanities and preserving the evidence' in Martin Paul Eve and Jonathan Gray (eds), Reassembling scholarly communications: histories, infrastructures, and global politics of open access (Cambridge, MA, 2020), p. 215.

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55. Ibid., p. 222.

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56. Coolahan, Women, writing, and language in early modern Ireland (Oxford, 2010); eadem, '"And this deponent further sayeth": orality, print and the 1641 depositions' in Marc Caball and Andrew Carpenter (eds), Oral and print cultures in Ireland, 1600–1900 (Dublin, 2009), pp 69–84; McAreavey, 'Re(-)membering women: Protestant women's victim testimonies during the Irish Rising of 1641' in Journal of the Northern Renaissance, ii (2010) (https://jnr2.hcommons.org/2010/1120/); eadem, "'This is that I may remember what passings that happened in Waterford": inscribing the 1641 Rising in the letters of the wife of the mayor of Waterford' in Early Modern Women: An Interdisciplinary Journal, v (fall 2010), pp 77–109; Joan Redmond, 'Memories of violence and New English identities in early modern Ireland' in Historical Research, lxxxix, no. 246 (Nov. 2016), pp 708–29; eadem, 'Religion, ethnicity and "conversion" in the 1641 Irish Rebellion' in The Seventeenth Century, xxxv, no. 6 (2020), pp 715–39.

56. 文献条目（作者名、题名、出版信息与网址保留原文以便检索）：Coolahan, Women, writing, and language in early modern Ireland (Oxford, 2010); eadem, ‘“And this deponent further sayeth”: orality, print and the 1641 depositions’ in Marc Caball and Andrew Carpenter (eds), Oral and print cultures in Ireland, 1600 – 1900 (Dublin, 2009), pp 69 – 84; McAreavey, ‘Re(-)membering women: Protestant women’s victim testimonies during the Irish Rising of 1641’ in Journal of the Northern Renaissance, ii (2010) (https://jnr2.hcommons.org/2010/1120/); eadem, “‘This is that I may remember what passings that happened in Waterford”: inscribing the 1641 Rising in the letters of the wife of the mayor of Waterford’ in Early Modern Women: An Interdisciplinary Journal, v (fall 2010), pp 77 – 109; Joan Redmond, ‘Memories of violence and New English identities in early modern Ireland’ in Historical Research, lxxxix, no. 246 (Nov. 2016), pp 708 – 29; eadem, ‘Religion, ethnicity and “conversion” in the 1641 Irish Rebellion’ in The Seventeenth Century, xxxv, no. 6 (2020), pp 715 – 39.

57. Catherine D'Ignazio and Lauren Klein, Data feminism (Cambridge, MA, 2020), p. 19.

57. 文献条目（作者名、题名、出版信息与网址保留原文以便检索）：Catherine D’Ignazio and Lauren Klein, Data feminism (Cambridge, MA, 2020), p. 19.

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